

Quality Management

1 — Introduction and the European framework

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BSc in Engineering Management • Università degli Studi di Perugia

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What this course is about

Five blocks, in this order

1. Regulations and the EU framework
2. History of quality management
3. Accreditation, certification and reliability
4. Statistical tools for quality management
5. Measurement techniques for quality control

The question behind all five

How do you convince someone who has never met you that your product does what you say it does?

Who is teaching

Prof. Gianluca Rossi

- Master in Mechanical Engineering, UnivPM
- Full Professor at UniPg since 2008, in Mechanical and Thermal Measurements
- Also teaches *Mechanical Measurement by processing waves and images*
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dr. Lorenzo Capponi

- BSc and MSc in Mechanical Engineering, UniPg
- PhD in Mechanical Measurement for Engineering and Space
- Postdoctoral Research Associate, University of Illinois (NASA-funded)
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ASK QUESTIONS

During the lecture, not after it. If something does not make sense, it is far more likely that it was explained badly than that you missed it.

Who else you will hear from

Three guest seminars — 15 October, 5 November, 26 November

Quality management is not an academic subject. It is what a company does on a Tuesday because a customer, a regulator or an auditor requires it.

Three sessions are given by people who do this for a living — in manufacturing, in aerospace supply, and in the national metrology system.

These count

What the guests present is part of the course and is examinable. They are also the clearest view you will get of how the rest of this material is actually used.

BRING A QUESTION

Come to each seminar with one question written down. “What does a normal week look like for you?” is a perfectly good one, and tells you more than any slide here will.

How the course works

When

- Monday 10:30–13:30, Room 3
- Thursday 11:30–13:30, Room 2
- 21 September – 3 December

Prerequisites

None.

Exam

Oral examination.

Material

- These slides, published after each lecture
- Your own notes
- *Juran's Quality Control Handbook* — reference, not required reading cover to cover

Everything is online

github.com/LolloCappo/student-hub — schedule, slides, and the notebooks for the hands-on session.

How to get the most out of this course

Said once, now, because it changes what you get in December

- **Take notes by hand, not on the slides.** The slides are published afterwards, so copying them down is wasted effort. Write the things I say that are *not* on the screen — that is where the exam questions come from.
- **Learn the vocabulary early.** Half of this subject is words used precisely. If “quality assurance” and “quality control” still feel interchangeable in November, the oral exam will be painful.
- **Connect it to something you own.** Every concept here applies to objects in your pocket. If you can find the example yourself, you will not need to memorise the definition.
- **Interrupt.** A three-hour lecture where nobody speaks is three hours I have got wrong.

QUESTION TO START WITH

Think of the last thing you bought that disappointed you. Was it badly *made*, or was it made exactly as intended and simply not what you expected? Hold that distinction — we come back to it all semester.

PART 1

The words we will use

Used loosely in industry, precisely in this course

Quality, and managing it

- **Quality** — conformance to requirements. Not “good”, not “expensive”, not “luxurious”: *conforming to what was specified*.
- **Quality Management** — the management activities and functions by which an organisation determines its quality policy and implements it, through quality planning, quality assurance and quality control.

WHY THE DEFINITION IS DELIBERATELY UNGLAMOROUS

A disposable syringe and a Formula 1 gearbox can both have perfect quality. They conform to what was specified. One costs 8 cents, the other a fortune. Quality is not the same axis as value.

AND WHY THAT MATTERS COMMERCIALY

If quality means “expensive”, then improving quality must raise cost, and quality becomes something you trade away under pressure. If quality means “conforming”, improving it *lowers* cost — because non-conformance is what you were paying for. We prove that number in Lecture 4.

The distinction that matters most

Quality Control and Quality Assurance

Quality Control (QC)

Checking that the output conforms.

A **corrective** tool, applied to the **product**.

Inspection, testing, measurement, validation, peer review.

Quality Assurance (QA)

Building confidence that requirements *will* be met.

A **preventive** tool, applied to the **process**.

Audits, procedures, checklists, training, methodology.

ONE SENTENCE TO REMEMBER IT BY

QC finds the defective bearing. QA is the reason the line stopped producing defective bearings.

A TEST YOU CAN APPLY ANYWHERE

Ask: *does this activity look at the product, or at the way the product is made?* Product → control. Process → assurance. Weighing every bag of flour is control. Writing the procedure that says how often to calibrate the scale is assurance.

Systems, processes, and time

- **Quality Management System (QMS)** — the whole structure: quality planning, improvement activities, a set of policies and objectives acting as guidelines within the organisation, with both QA and QC inside it. *This is what ISO 9001 certifies.*
- **Process** — any activity, or set of activities, that uses resources to transform inputs into outputs. Quality management acts on processes, not on people.
- **Reliability** — the ability to perform the required function, under stated conditions, *for a stated period of time.*

QUALITY IS AT DELIVERY; RELIABILITY IS OVER THE LIFETIME

A bolt meeting every dimension on the drawing the day it ships has **quality**. A bolt still holding after ten years of vibration and salt spray has **reliability**. Different properties, different measurements, different standards — and they can disagree.

“ACTS ON PROCESSES, NOT ON PEOPLE” IS NOT A PLATITUDE

If the same error recurs across three different operators, the operators are not the variable. The procedure, the tooling or the training is. Blaming the person is the fastest way to guarantee the error returns.

Quick exercise

QA or QC? Decide before I say.

1. Measuring the diameter of every tenth shaft
2. Training a new welder before they touch a production part
3. X-raying a casting for internal voids
4. Writing the work instruction for the X-ray operation
5. Auditing a supplier's factory before the first order
6. Rejecting a batch that failed the audit's criteria
7. Calibrating the gauge used for the measurement
8. A customer complaint arriving by email

THINK ABOUT NUMBER 8

A customer complaint is not QA or QC — it is the total failure of both, arriving from outside. It is also, uncomfortably, the cheapest quality data your company will ever receive. Why do most organisations handle it as a nuisance rather than as a measurement?

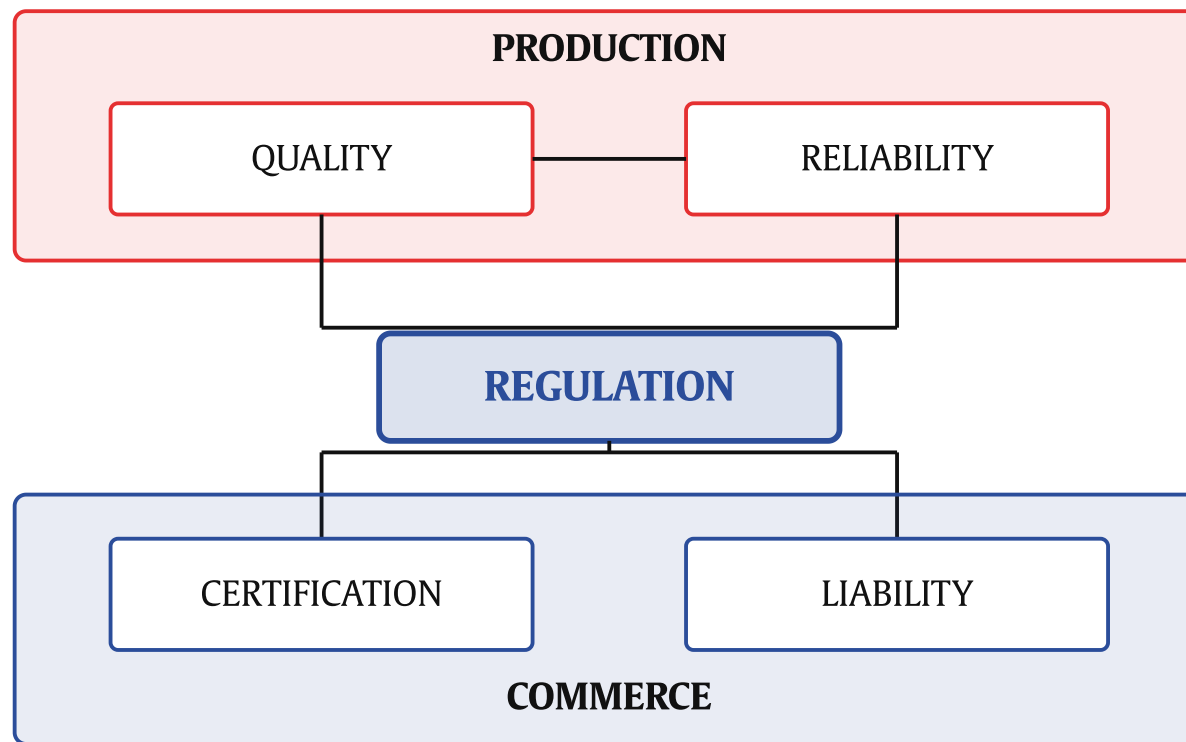
Recap **TAKEAWAYS**

Vocabulary — if only four things survive today, these four

- **1 Quality means conformance to requirements** — not goodness, not price. Both a syringe and a gearbox can have perfect quality.
- **2 QC looks at the product and corrects; QA looks at the process and prevents.** Product or process — that is the whole test.
- **3 The QMS is the container** that holds planning, policy, QA and QC together. ISO 9001 certifies the container.
- **4 Quality is at delivery, reliability is over time.** Required function, stated conditions, stated period — remove one and the word means nothing.

The map

Come back to this slide whenever you lose the thread



Regulation is the hinge between making something and selling it. Everything in this course hangs off one of these five boxes.

PART 2

The European framework

*Why a machine built in Perugia can be sold in
Hamburg without being tested twice*

The problem, before the solution

Europe, early 1980s

Every country wrote its own technical rules — for safety, for materials, for testing. Each rule was defensible on its own. Together they made a single European market impossible.

WHAT THAT COST

The same pressure vessel needed separate approval in France, Germany and Italy: three test campaigns, three sets of fees, three timelines — for one product. A small manufacturer simply did not export.

A national rule does not have to be protectionist to act as a trade barrier. It only has to be *different*.

CONSIDER

If you were a national regulator in 1980, would you have agreed to drop your own safety rules in favour of someone else's? On what condition?

What a technical barrier actually looks like

None of these was written to keep foreigners out

- **Different plugs and voltages** — an appliance certified in one country physically cannot be sold in another without redesign.
- **Different test methods for the same property** — two countries both require a flammability test, but specify different flames, durations and pass criteria. The product is safe in both; the paperwork is valid in neither.
- **Different approval bodies** — each state recognises only its own laboratory, so the same test is paid for repeatedly.
- **Different labelling and language requirements** — a separate print run per market for an identical product.

WHY THE “NOT PROTECTIONIST” POINT MATTERS

Each rule above has a genuine safety or consumer-information justification. That is precisely what makes the problem hard: you cannot solve it by banning bad faith. You have to build a mechanism that works even when everyone is acting reasonably.

The European Single Market

- Established **1 January 1993**
- Also called the Internal Market or the Common Market
- 27 EU member states, plus non-EU states through the EEA

The part that concerns engineers

Free movement of goods is achieved through *common rules and standards* that all participating states are legally committed to follow. That is where this course starts.

It guarantees the **four freedoms** — free movement of:

goods

capital

services

people

The stated purposes were economic: increase competition, labour specialisation and economies of scale, so that goods and factors of production move to where they are most valued — and drive integration of once-separate national economies into one.

Three kinds of barrier

All three had to go

Physical

Border checkpoints, customs queues, paperwork at the frontier.

Technical

National rules, standards and approvals that differ from country to country.

Fiscal

Value-added tax and excise duties applied differently across states.

Ours is the middle one

Physical barriers were removed politically, fiscal barriers fiscally. The **technical** barrier is the engineering problem — and the only one that could not be solved by a signature alone.

WHY THE TECHNICAL ONE IS HARDEST

You can abolish a customs post overnight. You cannot abolish the fact that a pressure vessel really does need testing. The rule has to be *replaced*, not removed — and replaced by something 27 states will each accept.

Recap **TAKEAWAYS**

The problem — before we look at the solution

- **1 A single market needs free movement of goods**, and that is one of the four freedoms established on 1 January 1993.
- **2 Three barrier types stood in the way** — physical, technical, fiscal. Only the technical one is an engineering problem.
- **3 A rule does not need bad intent to block trade.** It only needs to be different from the neighbouring country's.
- **4 So the rules could not simply be abolished.** They had to be replaced by something every member state would accept — which is the rest of this lecture.

How the EU actually legislates

Four instruments — and the difference is not a formality

- **Regulation** — binding in its entirety and **directly applicable** in every member state. It becomes law the day it applies, with no national act required.
- **Directive** — binding **as to the result** to be achieved, but each state chooses the form and method. It must be *transposed* into national law.
- **Decision** — binding only on those it is addressed to — a state, a company.
- **Recommendation & Opinion** — not binding at all.

WHY YOU MUST KNOW WHICH ONE YOU ARE READING

A **Directive** means you obey the *Italian* law that transposed it, and the detail may differ slightly from Germany's. A **Regulation** means the same text binds you and your German competitor identically. The CE marking rules sit in a Regulation; most product safety rules sit in Directives. That is why national implementations of product law still vary.

The information procedure

One idea, three directives, forty years

Rather than forbidding national rules, Europe made them **transparent**: a state must announce a draft technical regulation before adopting it.

1

83/189/EEC

1983 — the original directive

2

98/34/EC

1998 — codified, extended to information society services

3

(EU) 2015/1535

in force since 7 October 2015 — the one that applies today

Why it still matters

The directive has been replaced twice, but the mechanism has not changed since 1983. Transparency, not prohibition, is what stops national rules fragmenting the market.

How the procedure works

1

Notification

The member state sends the draft technical regulation to the Commission and to the other member states.

2

Standstill

Three months in which it may not be adopted. The Commission and other states may comment.

3

Decision

In light of the comments, the state adopts, amends or drops the rule.

ONE EXCEPTION

An urgency procedure allows immediate adoption in serious and unforeseeable circumstances — a public health or safety emergency. The Commission decides whether the urgency is justified; if it accepts, the three-month standstill does not apply.

THE QUIET POWER OF A WAITING PERIOD

Three months is short enough that a genuine rule survives it, and long enough that a rule designed to advantage a domestic producer gets noticed and challenged. Nobody has to prove bad faith — the delay does the work.

The New Approach, 1985

Council Resolution 85/C 136/01 — the idea that made it work

Harmonising every technical detail of every product in Europe by law was impossible. So the work was split in two.

Legislative harmonisation

Directives set only the *essential requirements* — mainly safety and health — that a product must meet. The law says *what*, never *how*. Meet them, and the product circulates freely.

Regulatory harmonisation

Standards bodies — CEN, CENELEC, ETSI — write the harmonised technical standards that say *how* to meet them. Voluntary, not law.

Why split it this way

Law is slow to change and hard to agree; technology is neither. Putting the *what* in law and the *how* in standards lets the detail be updated by engineers without reopening a political negotiation every time a material improves.

Presumption of conformity

The mechanism that makes a voluntary standard irresistible

- 1 The Commission **requests** a standard from CEN, CENELEC or ETSI
- 2 The standard is written and its reference is published in the **Official Journal**

- 3 Follow it, and you are **presumed** to meet the essential requirements it covers

You may still prove conformity another way. But then **the burden of proof is yours** — you must demonstrate, to an authority that did not ask for your opinion, that your alternative is equivalent.

VOLUNTARY IN THEORY, COMPULSORY IN PRACTICE

A harmonised standard is not law. But the choice is: follow it and the burden of proof lies with the authority, or ignore it and the burden lies with you. Almost every manufacturer follows it. That is how a voluntary document acquires the force of a rule without ever becoming one.

Mutual recognition

The principle

If a product is lawfully manufactured and marketed in one member state, there is no reason it should not be sold throughout the rest.

Member states may still issue their own mandatory technical rules where safety and health require it — but only if those rules comply with Community law and continue to guarantee free movement of goods. Policing that is exactly what the notification procedure does.

IN PRACTICE

Your machine is assessed once, in Italy, against a harmonised standard. It is then sold in Germany, Spain and Poland with no second assessment. That one sentence is worth more to a manufacturer than the whole rest of this lecture.

Recap **TAKEAWAYS**

The solution — four moving parts

- **1 Transparency, not prohibition.** States must notify draft technical rules and wait three months. 83/189/EEC → 98/34/EC → (EU) 2015/1535.
- **2 The New Approach splits the work.** Directives set essential requirements — the *what*. Standards bodies write the *how*.
- **3 Presumption of conformity** is the lever: follow the harmonised standard and the burden of proof shifts off you. That is why a voluntary standard is followed almost universally.
- **4 Mutual recognition** closes the loop: assessed lawfully once, sellable everywhere.

PART 3

Putting a product on the market

What a manufacturer actually has to do — and who checks

Conformity assessment

How much scrutiny depends on how much risk

EU product law does not treat every product the same. The applicable directive prescribes a **module** — a procedure — scaled to the hazard.

Low risk — internal production control

The manufacturer assesses the product themselves, compiles the technical file, and declares conformity. **No third party is involved at all.**

Higher risk — a notified body is involved

An independent body examines the design, or the type, or audits the production quality system — before the product may be placed on the market.

SAME MARKING, VERY DIFFERENT JOURNEYS

A desk lamp and an implantable medical device both end up carrying CE. Behind the lamp is a folder in a drawer. Behind the device is a notified body, an audited quality system and a design examination certificate. *The marking looks identical either way* — which is exactly why it is so widely misunderstood.

Notified bodies

Not regulators, not consultants

- Conformity assessment bodies **designated by a member state** and *notified* to the Commission — hence the name.
- Each is notified for **specific legislation and specific products**. A body notified for pressure equipment cannot certify a toy.
- They are listed in a public European database, so a customer can verify that the number on your documentation is real.
- Their competence is itself checked — usually through accreditation by the national accreditation body. *In Italy: ACCREDIA. We meet it next lecture.*

NOTE THE LAYERING

The manufacturer makes the product. The notified body checks the manufacturer. The accreditation body checks the notified body. Each layer exists because the layer below it has a commercial interest in the answer.

The CE marking

Regulation (EC) No 765/2008

- The manufacturer's **own declaration** that the product meets all applicable essential requirements
- It is *not* a quality mark, *not* a mark of origin, and *not* a certificate issued by an authority
- Where a notified body was required, its **four-digit number** appears beside the marking
- Affixing it without grounds is an offence across the member states

LOOK AROUND THE ROOM

Every laptop, phone charger and wall socket here carries it. It is the most widely reproduced engineering symbol in Europe, and almost nobody can say what it legally means.

Regulation 765/2008 and Decision 768/2008/EC together form the *New Legislative Framework*.

SPOT THE DIFFERENCE

If you see CE with a number next to it, what does that tell you that plain CE does not?

The EU Declaration of Conformity

The document behind the marking

The CE marking is the visible symbol. The **Declaration of Conformity** is the legal document it stands for, drawn up and signed by the manufacturer. It states:

- who the manufacturer is, and who is responsible
- which product it covers, identified unambiguously
- which EU legislation the product complies with
- which harmonised standards were applied
- the notified body and certificate number, where one was involved

Behind it sits the **technical file**: drawings, calculations, test reports, risk assessment — retained for years and producible on demand.

THIS IS WHERE QUALITY MANAGEMENT BECOMES LEGALLY UNAVOIDABLE

Someone signs that declaration. If the technical file does not support it, that signature is the problem. Everything we study for the rest of the semester — procedures, records, calibration, audits — exists so that the file *does* support it.

Who checks, after the fact

Market surveillance

Nobody inspects your product before it goes on sale — that is the whole point of self-declaration. Control happens **afterwards**, in the market.

- Each member state appoints **market surveillance authorities** with powers to sample products, demand the technical file, and require corrective action.
- They can order a product **withdrawn** from the market or **recalled** from users.
- An EU **rapid alert system** circulates notifications of dangerous non-food products between all member states, so a hazard found in one country is acted on in the others within days.

THE BARGAIN THIS REPRESENTS

Europe traded pre-market approval for post-market enforcement. You get to market fast and cheap; in exchange, if you were wrong, the consequences land on you publicly and across 27 countries at once.

When it goes wrong

The cost is not the fine

What happens

- withdrawal from sale, or recall from users
- national penalties, which vary by member state
- alert circulated across the EU
- product liability claims — Directive 85/374/EEC, which we meet next lecture

What it actually costs

- the logistics of retrieving units already sold
- the halted production line
- the customer who now audits you annually
- the reputation, which does not recover on the same timescale

The connection to this whole course

Quality management is not paperwork for its own sake. It is the mechanism that keeps you on the left-hand side of this slide.

Take out something you own

Phone charger, headphones, laptop — anything with a label

FIND AND ANSWER

1. Is there a **CE** marking? Is there a four-digit number next to it?
2. Who is named as the manufacturer, and in which country are they established?
3. If it was made outside the EU, who is the **importer** — and why does EU law insist that somebody inside the Union be named?
4. Which essential requirements would you guess apply to it — electrical safety, electromagnetic compatibility, restricted substances?

WHY QUESTION 3 IS THE INTERESTING ONE

If the manufacturer is in another continent, a European authority cannot easily reach them. Naming an importer or authorised representative inside the EU means there is always somebody within reach of the rules — somebody who can be asked for the technical file.

Where we got to **TAKEAWAYS**

Vocabulary, and the problem — 1 to 3 of 6

- **1 Quality is conformance to requirements.** QC corrects the product; QA prevents through the process.
- **2 A single market needs common rules,** because differing national technical rules fragment it — even when written in good faith.
- **3 Transparency, not prohibition:** notify the draft, wait three months.

Where we got to **TAKEAWAYS**

The solution, and the market — 4 to 6 of 6

- **1 The New Approach** puts essential requirements in law and the technical detail in harmonised standards — which carry presumption of conformity.
- **2 Risk sets the scrutiny:** self-declaration for low risk, a notified body for high risk. Same CE marking either way.
- **3 Control is post-market.** The manufacturer signs the Declaration of Conformity; surveillance authorities check afterwards.

Next time

Who writes these standards — ISO, IEC, CEN, CENELEC, UNI, CEI — and how quality, reliability, certification and liability fit together. Start from **the map**.

Questions?

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